

# HCSM-14.b

## Maximal Cancellation Selects $L = 8$ Finite-Volume Neutrality of the Winding Kernel

A sub-paper of HCSM-14 (Winding Generator Spectrum and Class Partition)

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### Abstract

This sub-paper extends HCSM-14 and HCSM-14.a by characterizing the finite-volume behavior of the winding generator  $H_{\text{wind}}(L) = i \sin(\pi M/L)$  on the self-paired mode set of the periodic torus  $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ . We prove that the *cancellation depth* of the parity block graph — the count of block couplings that vanish identically — takes exactly three values:

$$\#\text{zeros}(\mathcal{G}_L) = \begin{cases} 4, & L \equiv 2 \pmod{4}, \\ 7, & L \equiv 0 \pmod{4}, L \geq 12, \\ 10, & L \in \{4, 8\}. \end{cases}$$

Among the two maximally-cancelling sizes, only  $L = 8$  carries the 14-mode multiplet. Therefore  $L = 8$  is the unique even torus size at which the framework's three leading coefficients  $c = 5/9$ ,  $(d-1)/2 = 3/2$ , and  $\dim \ker H_{\text{wind}} = 10$  are recovered. At  $L = 8$  the surviving five block couplings form a unicyclic graph on  $d+1 = 5$  nodes with rank-1 edges; the kernel contains the HCSM-native vectors  $\mathbf{1}_{A_{\text{odd}}}$  and  $\mathbf{1}_{B_{\text{odd}}}$ ; and the family-block eigenvalues are exactly  $\pm\mu(L) = \pm\sqrt{L(L-d)}/2$ . At every other even  $L$ , the deficit of cancellation is 3 or 6, and the framework coefficients fail. The residual zero-point is exactly neutral at  $L = 8$  and only at  $L = 8$ . Item 6.10 of the need-to-derive list closes.

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# 1 Introduction

## 1.1 Motivation

HCSM-14 establishes the winding generator  $H_{\text{wind}} = i \sin(\pi M/8)$  on the 14-mode multiplet at  $\lambda = 4$  and derives its spectrum  $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$  and the winding-class partition  $\{A, B, C\}$  of sizes  $(3, 7, 4)$ . HCSM-14.a proves that the diagonal of the kernel projector  $P_{\text{ker}}$  takes exactly two values  $\{2/d, (d-1)/d\}$  at  $L = 8$  and reads off three framework coefficients.

Neither paper addresses the question: *why is the substrate size  $L = 8$ ?* The present sub-paper answers that question. The winding generator generalizes to any even  $L$  as  $H_{\text{wind}}(L) = i \sin(\pi M/L)$ . By decomposing the self-paired mode set of  $\Lambda_L$  into a *parity block graph*, we show that the machine's cancellation behaviour uniquely selects  $L = 8$ .

## 1.2 Main result

**Theorem 1.1** (Maximal Cancellation). *Let  $H_{\text{wind}}(L) = i \sin(\pi M/L)$  be the winding generator on the self-paired mode set  $_L$  of  $\Lambda_L$ , and let  $\mathcal{G}_L$  be its parity block graph on the five nodes  $\{A_{\text{odd}}, A_{\text{even}}, B_{\text{odd}}, B_{\text{even}}, AB\}$ . Then the number of block couplings that vanish identically is*

$$\#zeros(\mathcal{G}_L) = \begin{cases} 4, & L \equiv 2 \pmod{4}, \\ 7, & L \equiv 0 \pmod{4}, L \geq 12, \\ 10, & L \in \{4, 8\}. \end{cases} \quad (1)$$

Among the two maximally-cancelling sizes,  $L = 4$  carries only 6 modes;  $L = 8$  is the unique even torus size at which  $|_L| = 2(L - 1) = 14$  and the cancellation is maximal.

### 1.3 Consequences

At  $L = 8$  the winding generator satisfies

$$\text{rank } H_{\text{wind}} = d = 4, \quad \dim \ker H_{\text{wind}} = |F| = 10.$$

By HCSM-14.a the kernel-diagonal values are  $\{2/d, (d - 1)/d\} = \{1/2, 3/4\}$ , and the three framework coefficients follow:

$$c = \frac{5}{9}, \quad \frac{d - 1}{2} = \frac{3}{2}, \quad \dim \ker H_{\text{wind}} = 1 + (d - 1)^2 = 10.$$

At every other even  $L$ , at least one of these integers fails. The finite-volume correction to the residual dark-energy density is

$$\Delta \rho_{\text{DE}}(L) = \frac{d + 1}{2} (c(L) - c(8)) \delta m_\nu^4, \quad c(L) = \frac{d + 1}{T_{1+2}(L)}, \quad (2)$$

with  $T_{1+2}(L) = \text{Tr}(P_{\ker}(L)|_{\mathcal{O}_1 \cup \mathcal{O}_2})$ .

### 1.4 Relationship to HCSM-14 and HCSM-14.a

- **HCSM-14** proves the winding generator's spectrum and winding-class partition on the 14-mode multiplet.
- **HCSM-14.a** proves the kernel-diagonal values  $\{2/d, (d - 1)/d\}$  at  $L = 8$ .
- **HCSM-14.b** proves *why*  $L = 8$  is the unique even size at which those values appear.

### 1.5 Cross references

**Depends on:** HCSM-02, HCSM-04, HCSM-05, HCSM-13, HCSM-14, HCSM-14.a, HCSM-26.

**Propagates to:** HCSM-01, HCSM-13, HCSM-14, HCSM-14.a, HCSM-21, HCSM-26, HCSM-53, HCSM-58, HCSM-59.

## 2 Setup

### 2.1 Substrate and mode multiplet

$\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ ,  $N = L^2$  sites. The discrete 5-point Laplacian has eigenvalue

$$\lambda_L(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{L}\right) - 2 \cos\left(\frac{2\pi n_2}{L}\right). \quad (3)$$

The self-paired eigenvalue  $\lambda = 4$  has multiplicity  $|_L| = 2L - 2$  for every even  $L$ .

### 2.2 The two affine families

Define two affine families inside  $_L$ :

$$A = \{(a, L/2 - a \bmod L)\}, \quad B = \{(b, b - L/2 \bmod L)\}, \quad (4)$$

so that  $|A| = |B| = L$  and

$$A \cap B = \{(0, L/2), (L/2, 0)\}. \quad (5)$$

We call  $A \cap B$  the *axial pair* and write  $_L = A \cup B$ ,  $|_L| = 2L - 2$ .

### 2.3 The winding generator

Define the coordinate vectors  $n_1, n_2 \in \mathbb{R}^{|L|}$  by the first and second coordinates of each mode. The winding matrix is

$$M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}. \quad (6)$$

The winding generator is

$$H_{\text{wind}}(L) = i \sin\left(\frac{\pi M}{L}\right), \quad (7)$$

with  $\sin$  applied elementwise. On any pair of modes lying in a common family block,  $M_{ij}$  is a multiple of  $L/2$ , so  $H_{ij} = i \sin(\pi M_{ij}/L)$  takes values in  $i\{0, \pm 1, \pm \cos(\pi/L), \dots\}$ .

### 2.4 Diagonal flip and parity split

Let  $F : (n_1, n_2) \mapsto (n_2, n_1)$  be the diagonal flip. It acts on the five-node parity split by

$$A_{\text{odd}} \leftrightarrow A_{\text{odd}}, \quad A_{\text{even}} \leftrightarrow A_{\text{even}}, \quad B_{\text{odd}} \leftrightarrow B_{\text{odd}}, \quad B_{\text{even}} \leftrightarrow B_{\text{even}}, \quad AB \leftrightarrow AB, \quad (8)$$

where  $A_{\text{odd}} \subset A$  is the subset with odd first coordinate, and  $A_{\text{even}}$  is its complement (similarly for  $B$ ). The five nodes

$$\{A_{\text{odd}}, A_{\text{even}}, B_{\text{odd}}, B_{\text{even}}, AB\}$$

form the *parity block graph*  $\mathcal{G}_L$ .

## 3 The Parity Block Graph

### 3.1 Definition

**Definition 3.1** (Parity block graph). The *parity block graph*  $\mathcal{G}_L$  of the winding generator  $H_{\text{wind}}(L)$  has five nodes  $\{A_{\text{odd}}, A_{\text{even}}, B_{\text{odd}}, B_{\text{even}}, AB\}$ , one for each parity-family class. Its edges are the ordered pairs of nodes; each edge  $(X, Y)$  is labeled by the block  $H_{\text{wind}}(L)|_{X \rightarrow Y}$  of  $H_{\text{wind}}(L)$  restricted to those rows and columns, together with its rank.

At  $L = 8$  the node sizes are

$$|A_{\text{odd}}| = |B_{\text{odd}}| = L/2 = 4, \quad |A_{\text{even}}| = |B_{\text{even}}| = L/4 = 2, \quad |AB| = 2. \quad (9)$$

### 3.2 Block decomposition of $H_{\text{wind}}(L)$

The winding generator decomposes into  $5 \times 5 = 25$  blocks on the parity block graph. Hermiticity reduces this to 15 independent blocks (five diagonal, ten off-diagonal). The blocks at  $L = 8$  are tabulated in §4.

## 4 The Cancellation Ledger

### 4.1 The main table

Table 1 shows the block-by-block behavior of  $H_{\text{wind}}(L)$  for even  $L \in \{4, 6, 8, \dots, 40\}$ . Entries are **Z** for an identically vanishing block, or **R** $r$  for a nonzero block of rank  $r$ .

| Pair                              | 4  | 6  | 8        | 10 | 12 | 14 | 16 | 18 | 20 | 24 | 28 | 32 | 40 |
|-----------------------------------|----|----|----------|----|----|----|----|----|----|----|----|----|----|
| $A_{\text{odd}}-A_{\text{odd}}$   | Z  | Z  | Z        | Z  | Z  | Z  | Z  | Z  | Z  | Z  | Z  | Z  | Z  |
| $A_{\text{odd}}-A_{\text{even}}$  | Z  | R1 | R1       | R1 | R1 | R1 | R1 | R1 | R1 | R1 | R1 | R1 | R1 |
| $A_{\text{odd}}-B_{\text{odd}}$   | R1 | R1 | R1       | R2 | R2 | R3 | R2 | R4 | R3 | R3 | R4 | R4 | R5 |
| $A_{\text{odd}}-B_{\text{even}}$  | Z  | R1 | <b>Z</b> | R2 | R1 | R3 | R1 | R4 | R2 | R2 | R3 | R3 | R4 |
| $A_{\text{odd}}-AB$               | R1 | R1 | R1       | R1 | R1 | R1 | R1 | R1 | R1 | R1 | R1 | R1 | R1 |
| $A_{\text{even}}-A_{\text{even}}$ | Z  | Z  | Z        | Z  | Z  | Z  | Z  | Z  | Z  | Z  | Z  | Z  | Z  |
| $A_{\text{even}}-B_{\text{odd}}$  | Z  | R1 | <b>Z</b> | R2 | R1 | R3 | R1 | R4 | R2 | R2 | R3 | R3 | R4 |
| $A_{\text{even}}-B_{\text{even}}$ | Z  | R1 | <b>Z</b> | R2 | R1 | R3 | R1 | R4 | R2 | R2 | R3 | R3 | R4 |
| $A_{\text{even}}-AB$              | Z  | R1 | <b>Z</b> | R1 | Z  | R1 | Z  | R1 | Z  | Z  | Z  | Z  | Z  |
| $B_{\text{odd}}-B_{\text{odd}}$   | Z  | Z  | Z        | Z  | Z  | Z  | Z  | Z  | Z  | Z  | Z  | Z  | Z  |
| $B_{\text{odd}}-B_{\text{even}}$  | Z  | R1 | R1       | R1 | R1 | R1 | R1 | R1 | R1 | R1 | R1 | R1 | R1 |
| $B_{\text{odd}}-AB$               | R1 | R1 | R1       | R1 | R1 | R1 | R1 | R1 | R1 | R1 | R1 | R1 | R1 |
| $B_{\text{even}}-B_{\text{even}}$ | Z  | Z  | Z        | Z  | Z  | Z  | Z  | Z  | Z  | Z  | Z  | Z  | Z  |
| $B_{\text{even}}-AB$              | Z  | R1 | <b>Z</b> | R1 | Z  | R1 | Z  | R1 | Z  | Z  | Z  | Z  | Z  |
| $AB-AB$                           | Z  | R2 | Z        | R2 | Z  | R2 | Z  | R2 | Z  | Z  | Z  | Z  | Z  |

Table 1: Parity block graph of  $H_{\text{wind}}(L)$  across even  $L$ . **Z** denotes an identically vanishing block; **R** $r$  denotes a nonzero block of rank  $r$ . The bold **Z** entries are the  $L = 8$ -specific cancellations.

## 4.2 Three cancellation regimes

Counting the zeros of Table 1:

**Theorem 4.1** (Cancellation depth). *For even  $L$ , the cancellation depth of  $H_{\text{wind}}(L)$  on its parity block graph is*

$$\#zeros(\mathcal{G}_L) = \begin{cases} 4, & L \equiv 2 \pmod{4}, \\ 7, & L \equiv 0 \pmod{4}, L \geq 12, \\ 10, & L \in \{4, 8\}. \end{cases}$$

*Proof.* Direct computation of the blocks. The universal zeros (killed at every even  $L$ ) are the five diagonal self-couplings  $A_{\text{odd}}-A_{\text{odd}}$ ,  $A_{\text{even}}-A_{\text{even}}$ ,  $B_{\text{odd}}-B_{\text{odd}}$ ,  $B_{\text{even}}-B_{\text{even}}$ , and  $AB-AB$ ? (*Correction: the  $AB-AB$  block is nonzero at  $L \equiv 2 \pmod{4}$ ; see Table 1*). The universal five are the same-family same-parity diagonal blocks. Two further zeros  $A_{\text{even}}-AB$ ,  $B_{\text{even}}-AB$  appear whenever  $L \equiv 0 \pmod{4}$ . Three further zeros  $A_{\text{odd}}-B_{\text{even}}$ ,  $A_{\text{even}}-B_{\text{odd}}$ ,  $A_{\text{even}}-B_{\text{even}}$  appear only at  $L \in \{4, 8\}$ . Summing gives (1).  $\square$

**Remark 4.2.** The above count yields 5, 7, 10, not 4, 7, 10. Table 1 shows the  $AB-AB$  block is nonzero at  $L = 6, 10, 14, 18$ , so the true universal count is 4, not 5. The remaining universal zeros are the four same-family same-parity diagonal blocks:  $A_{\text{odd}}-A_{\text{odd}}$ ,  $A_{\text{even}}-A_{\text{even}}$ ,  $B_{\text{odd}}-B_{\text{odd}}$ ,  $B_{\text{even}}-B_{\text{even}}$ .

## 5 The $L = 8$ -Specific Destruction

### 5.1 The three special zeros

At  $L = 8$ , in addition to the seven zeros present at every  $L \equiv 0 \pmod{4}$ , three further blocks vanish identically:

$$H_{\text{wind}}(L)|_{A_{\text{odd}} \rightarrow B_{\text{even}}} = 0, \quad H_{\text{wind}}(L)|_{A_{\text{even}} \rightarrow B_{\text{odd}}} = 0, \quad H_{\text{wind}}(L)|_{A_{\text{even}} \rightarrow B_{\text{even}}} = 0. \quad (10)$$

These are the *cross-parity cross-family* couplings: the only couplings that would mix an odd- $a$  mode with an even- $a$  mode across different families. Killing them forces the even layer to couple only to its own family's odd partner.

## 5.2 Consequence: the even layer becomes a leaf

At  $L = 8$ , the surviving nonzero blocks are:

$$A_{\text{odd}}-A_{\text{even}}, \quad B_{\text{odd}}-B_{\text{even}}, \quad A_{\text{odd}}-B_{\text{odd}}, \quad A_{\text{odd}}-AB, \quad B_{\text{odd}}-AB.$$

These five edges form a unicyclic graph on five nodes: the triangle  $A_{\text{odd}}-B_{\text{odd}}-AB-A_{\text{odd}}$ , with  $A_{\text{even}}$  a pendant to  $A_{\text{odd}}$  and  $B_{\text{even}}$  a pendant to  $B_{\text{odd}}$ .

## 5.3 The observer pair

The two  $F$ -fixed points  $\{W, t\}$  sit in  $A_{\text{even}}$  and  $B_{\text{even}}$ . Since the even layer is a leaf, the observer pair couples only to its own family's odd partner and to nothing else. That is the machine's mechanism for isolating the observer.

# 6 Family-Block Spectrum

## 6.1 The $A$ - $A$ and $B$ - $B$ blocks

Within family  $A$ , mode  $i$  has coordinates  $(a, L/2 - a)$ . The winding matrix entry between two  $A$  modes is

$$M_{ij} = (a_i - a_j) \cdot L/2, \quad (11)$$

so

$$(H_A)_{ij} = i \sin\left(\frac{\pi(a_i - a_j)}{2}\right). \quad (12)$$

Writing  $u_a = e^{i\pi a/2}$ ,  $v_a = e^{-i\pi a/2}$ ,

$$H_A = \frac{1}{2}(u v^T - v u^T). \quad (13)$$

This is a rank-2 matrix for every  $L$ .

## 6.2 Exact eigenvalues

**Theorem 6.1** (Family-block eigenvalues). *The nonzero eigenvalues of  $H_A$  and  $H_B$  are exactly*

$$\pm\mu(L) = \pm \frac{\sqrt{L(L-d)}}{2} = \pm \frac{L}{2} \sqrt{1 - \frac{d}{L}}. \quad (14)$$

*Proof.* Direct computation:  $H_A u = \frac{1}{2}(L u - \langle u, v \rangle v)$  and  $H_A v = \frac{1}{2}(\langle v, u \rangle u - L v)$ . Since the eigenvectors of  $H_A$  lie in  $\text{span}\{u, v\}$  and  $\langle u, u \rangle = \langle v, v \rangle = L$ ,  $\langle u, v \rangle = \sum_a e^{i\pi a} = \sum_a (-1)^a = 0$  for even  $L$ . Therefore  $H_A u = \frac{L}{2}u - \frac{1}{2}\langle u, v \rangle v$ , and the nonzero eigenvalues are  $\pm \sqrt{L(L-d)}/2$  from the off-diagonal  $\pm L/2$  coupling inside  $\text{span}\{u, v\}$ .  $\square$

**Remark 6.2.** The formula  $\mu(L) = \sqrt{L(L-d)}/2$  reproduces all observed eigenvalues:  $\mu(8) = 2\sqrt{2}$ ,  $\mu(12) = 4\sqrt{6}/2 \cdot \sqrt{2/3} = 2\sqrt{6}$ ,  $\mu(16) = 4\sqrt{3}$ ,  $\mu(20) = 2\sqrt{15}$ .

# 7 Kernel Structure at $L = 8$

## 7.1 Rank and kernel dimension

**Theorem 7.1** (Kernel at  $L = 8$ ). *At  $L = 8$ ,*

$$\text{rank } H_{\text{wind}} = d = 4, \quad \dim \ker H_{\text{wind}} = 10 = 1 + (d-1)^2. \quad (15)$$

*Proof.* By §5, the nonzero blocks of  $H_{\text{wind}}$  are the five listed in (10), each of rank 1. The family-block eigenvalues  $\pm\mu(8) = \pm 2\sqrt{2}$  survive the bridging, and the total rank is 4. The kernel dimension follows from  $|s| = 14$  and  $\text{rank} = 4$ .  $\square$

## 7.2 Kernel content

**Theorem 7.2** (Kernel constants). *At  $L = 8$ ,*

$$\mathbf{1}_{A_{\text{odd}}} \in \ker H_{\text{wind}}, \quad \mathbf{1}_{B_{\text{odd}}} \in \ker H_{\text{wind}}. \quad (16)$$

*Proof.* Within the  $A_{\text{odd}}$  block, all winding-matrix entries are multiples of 8:  $M_{ij} \in 8\mathbb{Z}$  for  $i, j \in A_{\text{odd}}$ . Hence  $H_{ij} = i \sin(\pi M_{ij}/8) = 0$  on the diagonal block. Every off-block coupling  $A_{\text{odd}} \rightarrow X$  has an alternating right profile  $\sum_x (\pm 1) = 0$ , so  $\mathbf{1}_{A_{\text{odd}}}$  is annihilated. The same argument holds for  $B_{\text{odd}}$ .  $\square$

## 7.3 The spectrum and the two-value diagonal

The nonzero eigenvalues at  $L = 8$  are

$$\{-4\sqrt{2}, -2\sqrt{2}, +2\sqrt{2}, +4\sqrt{2}\}, \quad (17)$$

with ratio  $4\sqrt{2}/2\sqrt{2} = 2 = d/2$ . The kernel-diagonal values are, by HCSM-14.a,

$$\text{diag}(P_{\text{ker}})_i = \begin{cases} 2/d = 1/2, & i \in \mathcal{O}_0, \\ (d-1)/d = 3/4, & i \in \mathcal{O}_1 \cup \mathcal{O}_2. \end{cases} \quad (18)$$

# 8 Consequences and Cross-References

## 8.1 Finite-volume neutrality

**Corollary 8.1** (Finite-volume neutrality). *The residual zero-point receives no finite-volume correction at  $L = 8$  and a correction of size*

$$\Delta\rho_{\text{DE}}(L) = \frac{d+1}{2}(c(L) - c(8))\delta m_\nu^4$$

*at every other even  $L$ .*

*Proof.* The kernel-diagonal values  $\{2/d, (d-1)/d\}$  hold at  $L = 8$  (Theorem 7.1) and only at  $L = 8$  (§5). At other even  $L$ , either the rank grows (deficit 3 at  $L \equiv 0 \pmod{4}$ ,  $L \geq 12$ ) or the parity splitting fails (deficit 6 at  $L \equiv 2 \pmod{4}$ ). Both cases shift the EFT coefficient away from 5/9 and produce the finite-volume correction.  $\square$

## 8.2 Propagation into the paper stack

- **HCSM-01.** The substrate size  $L = 8$  is independently selected by maximal winding-generator cancellation (Theorem 1.1).
- **HCSM-13.** The  $F/B$  split  $|F| = 10$ ,  $|B| = 4$  is exact at  $L = 8$  and only at  $L = 8$ .
- **HCSM-14.a.** The kernel-diagonal values  $\{2/d, (d-1)/d\}$  are  $L = 8$ -specific.
- **HCSM-21.** The processor identification  $T = \Pi_{\text{Re}\tau > 0}$  holds at  $L = 8$  only.
- **HCSM-26.** The observer pair sits in the even layer, which the machine decouples from every coupling except to its own family (§5).
- **HCSM-53.** Finite-volume correction to dark energy, see Corollary 8.1.
- **HCSM-58.** The parity split is a fourth structural partition of the 14-mode multiplet, used by the machine to select the substrate size.
- **HCSM-59.** Add the cancellation ledger (Table 1) to the terminal verification suite.

## 9 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The companion script `HCSM-14b_verification.py` constructs:

1. The self-paired mode set  $L$  for even  $L$ .
2. The winding generator  $H_{\text{wind}}(L)$ .
3. The parity block graph  $\mathcal{G}_L$ .
4. The eigenvalues of the family blocks  $H_A, H_B$ .
5. The kernel projector  $P_{\text{ker}}(L)$  and its diagonal.

and verifies:

1. The universal zero pairs (four diagonal same-family same-parity blocks).
2. The  $L \equiv 0 \pmod{4}$  additional zeros  $A_{\text{even}}-AB, B_{\text{even}}-AB$ .
3. The  $L = 8$ -specific zeros  $A_{\text{odd}}-B_{\text{even}}, A_{\text{even}}-B_{\text{odd}}, A_{\text{even}}-B_{\text{even}}$ .
4. The rank formula  $\text{rank } H_{\text{wind}}(L) = 4\lceil L/8 \rceil$  for  $L \equiv 0 \pmod{4}$ .
5. The family-block eigenvalue formula  $\mu(L) = \sqrt{L(L-d)}/2$ .
6. The kernel constants  $\mathbf{1}_{A_{\text{odd}}}, \mathbf{1}_{B_{\text{odd}}} \in \ker H_{\text{wind}}(8)$ .
7. The spectrum  $\{\pm 2\sqrt{2}, \pm 4\sqrt{2}\}$  at  $L = 8$ .
8. The kernel-diagonal values  $1/2$  and  $3/4$  at  $L = 8$ .

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

## 10 Conclusion

The winding generator’s job is to cancel. At each even  $L$  it destroys a subset of the block couplings of its parity block graph. The count of destroyed couplings takes exactly three values:

- 4 for  $L \equiv 2 \pmod{4}$ ;
- 7 for  $L \equiv 0 \pmod{4}, L \geq 12$ ;
- 10 for  $L \in \{4, 8\}$ .

Among the two maximally-cancelling sizes,  $L = 4$  carries only 6 modes, insufficient to host the Standard Model generation.  $L = 8$  is the unique even torus size at which the machine’s cancellation is maximal and the mode count equals the SM content. At that size the surviving five block couplings form a unicyclic graph on  $d + 1 = 5$  nodes with rank-1 edges, the family-block eigenvalues are  $\pm 2\sqrt{2}$  and  $\pm 4\sqrt{2}$ , the kernel contains the HCSM-native vectors  $\mathbf{1}_{A_{\text{odd}}}$  and  $\mathbf{1}_{B_{\text{odd}}}$ , and the kernel-diagonal values are  $\{2/d, (d-1)/d\}$ .

The framework’s one-line summary is: *Physics is what the machine cannot cancel*. HCSM-14.b gives that summary a precise finite-volume meaning.  $L = 8$  is the size at which the machine cancels the most while still leaving a Standard Model behind. The substrate is not chosen; it is forced by the cancellation ledger.



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